Two-Robot Assembly of a Sheet-Metal Frame with xArm6 – Technical Assignment

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Summary

- **Problem:** two xArm6 robots must assemble four U-profiles with tolerances of ±0.05 mm, while the robot repeatability is ±0.1 mm.
- **Key idea:** the robots only transport the parts; the fixture gives the accuracy.
- **Cell:** the robots face each other at 400 mm from the fixture. A Python script based on the official kinematic parameters shows that all pick and place points are reachable at 43–61% of the arm extension.
- **Gripper:** xArm Gripper with custom fingers that centre the profile and grip it at its centre. The robot releases the part before the fixture centres it.
- **Sequence and control:** R1 places both Pos 1, R2 both Pos 2. A PLC runs the sequence as an SFC; a Petri net proves that only one robot is in the fixture at a time.
- **Tolerances:** self-centring units, end stops, pushers, a gauge block and swing clamps. Error budget for the 120 mm dimension: ±0.021 mm (RSS), ±0.047 mm (worst case).
- **Parts:** the fixture cannot correct the height variation of the parts, so the operator loads measured kits with matched heights.
- **Safety and cycle time:** light fence with pull-out drawers for loading; about 40 s per assembly.
- **Validation:** a requirements–verification matrix and three checks: fixture measured once, first assemblies fully measured, feeler gauge on every assembly.

1 Introduction

ETH asked me two questions about drawing AMZ-SA-26-001: how I would set up and solve the pick-and-place task with two UFACTORY xArm6 robots, and how the drawing tolerances can be met when the robots alone are not accurate enough. Chapter 3 answers the first question, Chapter 4 the second.

I followed one method through the whole work: model, check, validate. I modelled the robot with its official kinematic parameters and the fixture with an error budget. I checked the layout numerically with a Python script. Then I planned how to validate the real cell with measurements. A full 3D simulation of the cell is a next step (Chapter 5).

Scope and assumptions.

- The parts arrive in trays, loaded by an operator.
- The robot task ends with the four parts located and clamped in the fixture. Joining is a separate step.
- The cell is for a small lab batch, not for series production.
- Values marked "assumed" are my estimates. All other data come from the drawing or from the UFACTORY documentation.

I sent two questions to ETH, on the meaning of "2 ×" and on the joining method. I had no answer at the time of writing, so I state my assumptions in Chapter 2 and base the whole work on them.

2 Drawing analysis

Drawing AMZ-SA-26-001 shows a frame of four sheet-metal U-profiles. The critical requirements are the assembly dimensions at ±0.05 mm and the geometric tolerances of 0.05 mm, while the single parts are made to ISO 2768-m, with tolerances up to ten times larger.

Parts. All profiles are S235JR cold-formed U-profiles (EN 10162), placed with the legs down.

Item	Profile (mm)	Length (mm)	Qty	Position
Pos 1	U 40 × 20 × 2	320	2	Bottom, along X
Pos 2	U 30 × 15 × 2	280	2	Top, along Y, resting on both Pos 1

Assembly requirements.

Requirement	Value
Distance between Pos 1 axes	120 ±0.05 mm
Distance between Pos 2 axes	120 ±0.05 mm
Pos 1 left end to Pos 2 axis (2 ×)	100 ±0.05 mm
Pos 1 outer edge to Pos 2 end (2 ×)	60 ±0.05 mm
Parallelism Pos 1 to Pos 1	0.05 mm
Perpendicularity Pos 2 to Pos 1	0.05 mm
Parallelism of the Pos 2 top face to the base plate	0.05 mm
Gap at the 4 contacts	≤ 0.05 mm

Part tolerances. The lengths have no written tolerance, so ISO 2768-m applies: 320 and 280 ±0.5 mm, 40 ±0.3 mm, 30, 20 and 15 ±0.2 mm.

What this means for the cell.

1. **Locate on axes, not on nominal edges.** The assembly tolerances (±0.05 mm) are much tighter than the part tolerances (up to ±0.5 mm). So the fixture must find the axis of each part by centring, and refer to the real edges of the parts already placed (Chapter 4).
2. **Part height affects parallelism.** The Pos 1 height is 20 ±0.2 mm. Pos 2 rests on both Pos 1, so a height difference tilts it. The fixture cannot correct this, so I measure and pair the parts (Section 4.4).
3. **Assumption 1: reading of "2 ×".** If 2 × 100 means 100 mm from both ends of the same Pos 1, the chain 100 + 120 + 100 fixes the part length at 320 mm. Each of the three dimensions has ±0.05 mm, so the length would need to be within 320 ±0.15 mm. The drawing allows 320 ±0.5 mm, so many parts would not fit. For example, a part of 320.30 mm is within its tolerance, but then at least one of the three dimensions would be 0.20 mm off instead of 0.05 mm. I therefore assume that 2 × 100 means 100 mm on each Pos 1, measured from their common left end.
4. **Assumption 2: joining.** The drawing does not say how the parts are joined. I assume that the robot task ends with the parts located and clamped in the fixture, and that joining is a separate step.

I sent the questions on points 3 and 4 to ETH.

3 Setting up and solving the pick-and-place task

The two xArm6 face each other across one fixture. R1 places both Pos 1, R2 places both Pos 2, and a PLC coordinates the two robots. One assembly takes about 40 s.

3.1 Cell layout and reachability check

The fixture centre is the origin of the cell. R1 stands at (0, −400) mm and R2 at (0, +400) mm. The trays are on the east side, on pull-out drawers loaded by the operator (Section 3.5). The west side stays free for unloading and joining.

I chose this layout because the place points are at 340–460 mm from the robot base, the robots approach the fixture from opposite sides, and R2 can wait on its own side while R1 works.

To check the layout with numbers, I wrote a Python script (reach_check.py, in the supporting material). It uses the official DH parameters of the xArm6, which I cross-checked against the UFACTORY ROS kinematics file: the two models differ by less than 0.003 mm over 1000 random poses. For each pick and place point, the script solves the inverse

kinematics with the gripper pointing down and computes the arm extension (shoulder-to-wrist distance divided by its maximum).

As a design criterion, I keep the extension between 30% and 75% (about 250–550 mm from the base). This keeps a margin from the singularities of a folded or fully stretched arm. It is my choice, not a UFACTORY limit.

Point	Distance from base (mm)	Arm extension
R1 places Pos 1, near	340	43%
R1 places Pos 1, far	460	61%
R2 places Pos 2, left and right	404	53%
R1 and R2 pick from the tray	400	52%

All six points are reachable and inside the criterion. Assumptions: tool length 200 mm, fixture plate 20 mm. The check covers geometry only; collisions and dynamics would need a 3D simulation, for example in ROS.

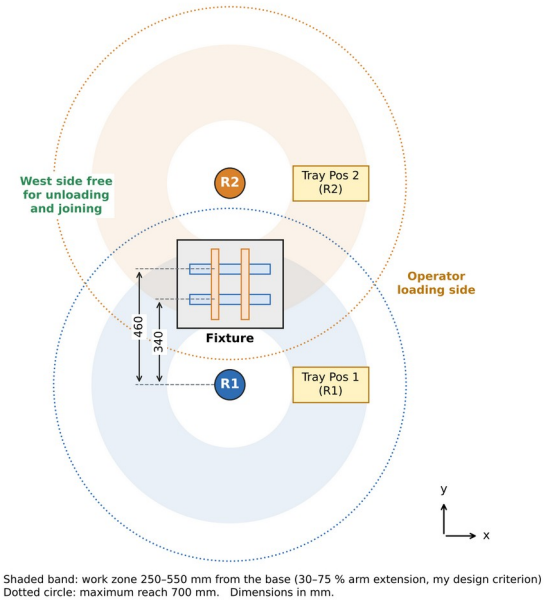


Figure 3.1 – Cell layout (top view, mm). R1 places Pos 1, R2 places Pos 2; the trays are on the east side.

3.2 Gripper and grasping

I use the xArm Gripper (stroke 84 mm, gripping force 30 N, Hardware Manual) with custom fingers. The fingers close on the two side walls (legs) of the U-profile from outside. A small shoulder on each finger rests on top of the profile (the web), so the part always sits at the same height in the gripper. Since the fingers close symmetrically, they also centre the part across its width during the grasp.

One gripper fits both profiles (40 and 30 mm wide), so both robots use the same tool. A vacuum gripper holds the part wherever the cup touches it, so it does not centre the part during the grasp. A magnetic gripper has the same problem and leaves residual magnetism on the steel.

The part is held by friction: $2 \times \mu \times N = 2 \times 0.2 \times 30 = 12$ N, against a Pos 1 weight of 3.7 N ($\mu = 0.2$ assumed). This allows moderate accelerations. Gripper and part weigh about 1.2 kg, well below the 5 kg payload.

I grip each profile at its centre, with fingers about 40–60 mm long. The ends must stay free for the centring units and the end stops. For Pos 2, the centre lies in the 80 mm gap between the two Pos 1, so the fingers do not touch them. The robot releases the part before the fixture centres it (Section 4.1).

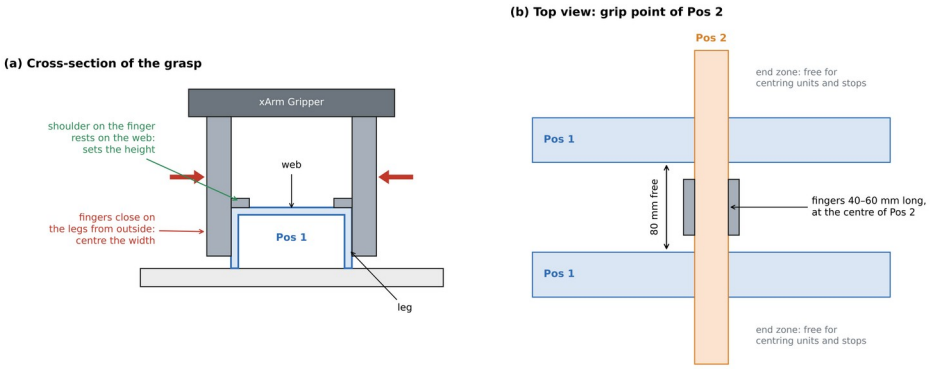


Figure 3.2 – Grasp of the U-profile: (a) cross-section; (b) grip point of Pos 2, between the two Pos 1.

3.3 Task allocation and sequence

I split the work by part type: R1 places both Pos 1, R2 both Pos 2. For a lab batch, simplicity is more important than cycle time. Each robot has one tray and one part type, and the fixture changes hands only once per cycle.

1. R1 places Pos 1a. The fixture centres it and pushes it against the end stop.
2. R1 places Pos 1b in the same way, returns home and releases the fixture.
3. Meanwhile, R2 picks Pos 2a and waits outside the fixture.
4. R2 places Pos 2a. The fixture centres it, pushes it against the 60 mm gauge block, then clamps it.
5. R2 places Pos 2b in the same way. The assembly is ready for joining.

Each part is located right after it is placed, so it is fixed before the robot comes back with the next one. For series production, splitting the work by side would shorten the cycle.

3.4 Control logic

A PLC supervises the cell and runs the sequence as an SFC (IEC 61131-3). Each robot runs its own motion programs. They exchange signals with the PLC through the digital I/O of their controllers (8 inputs and 8 outputs each).

The SFC has two parallel branches, one per robot, which join before R2 enters the fixture. Every transition uses a real signal: gripper position, centring and clamp sensors, drawer sensor. Every transition has a timeout that leads to an error step.

The critical rule is that only one robot may be in the fixture at a time. I modelled it with a Petri net: places "R1 out", "R1 in", "R2 out", "R2 in" and "fixture free". The place invariant *free + R1 in + R2 in = 1* holds for every transition, so at most one robot is in the fixture in every reachable state. I confirmed it with the invariant analysis of the PIPE tool. The SFC gives the order; the Petri net proves the exclusion.

Error handling has three levels. Recoverable errors (a centring unit not closed, a part not detected in the gripper) get one automatic retry, then the cell stops. Critical errors (part lost during transport, second failure, inconsistent sensors) stop the cell and call the operator. Emergency stop and collision are handled by the safety system and are never retried.

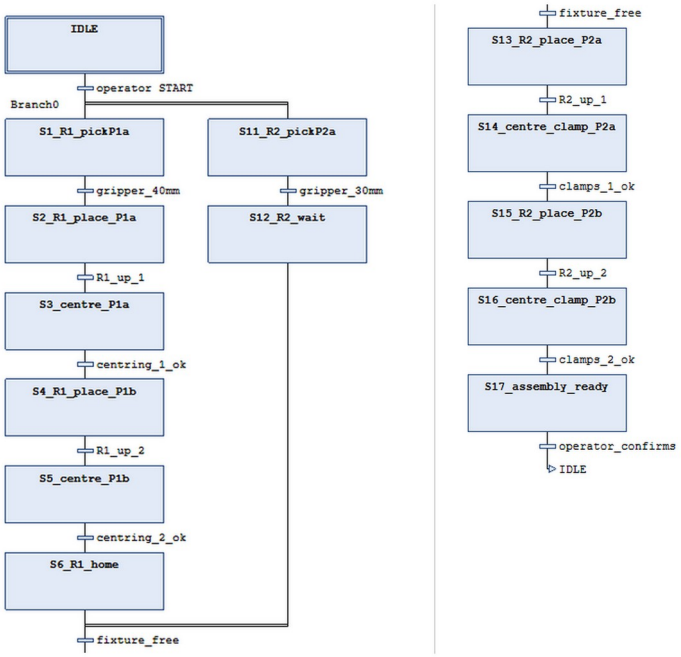


Figure 3.3 – SFC of the cell, drawn in Schneider SoMachine (IEC 61131-3). Two parallel branches join before R2 enters the fixture; the chart continues in the right column after fixture_free. Error handling (timeouts, part lost, emergency stop) is managed outside the chart.

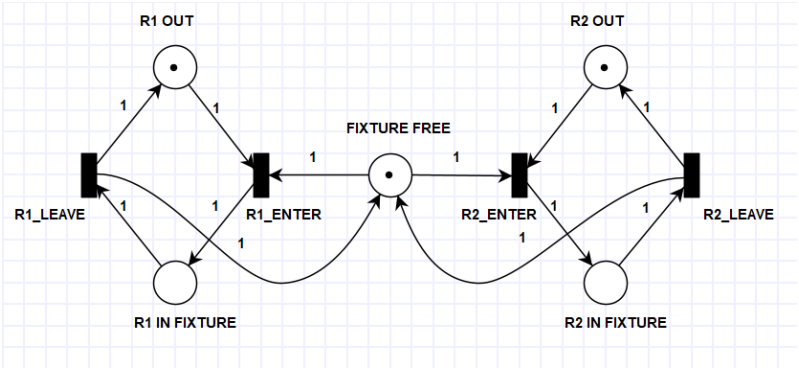


Figure 3.4 – Petri net of the mutual exclusion on the fixture (modelled and analysed in PIPE).

3.5 Safety and cycle time

A cobot does not make the application safe by itself: the whole cell needs a risk assessment (ISO 10218, ISO/TS 15066). The main hazards are the sheet-metal edges, pinching between the moving parts of the fixture, and an operator loading parts inside the robot reach.

I place a light fence outside the reach of the robots (700 mm) with one pull-out drawer per robot. The operator loads the measured kit with the drawer out, without entering the cell. A sensor confirms that the drawer is in before the robot picks, and the back wall of the drawer closes the opening when it is out. Both robots share one emergency stop, collision detection stays on, and the pneumatic pressure of the fixture is limited.

I estimate a cycle time of about 37 s per assembly: about 20 s for the R1 phase and 18 s for the R2 phase. I round it up to 40 s, because the estimate assumes straight moves without waiting times. I assume a transfer speed of 250 mm/s (25% of the 1 m/s maximum) and about 1 s for each approach, retreat, gripper, centring and clamping action. I keep the speed low because the part is held by friction and for safety. The real value must be measured on the cell (details in the Appendix).

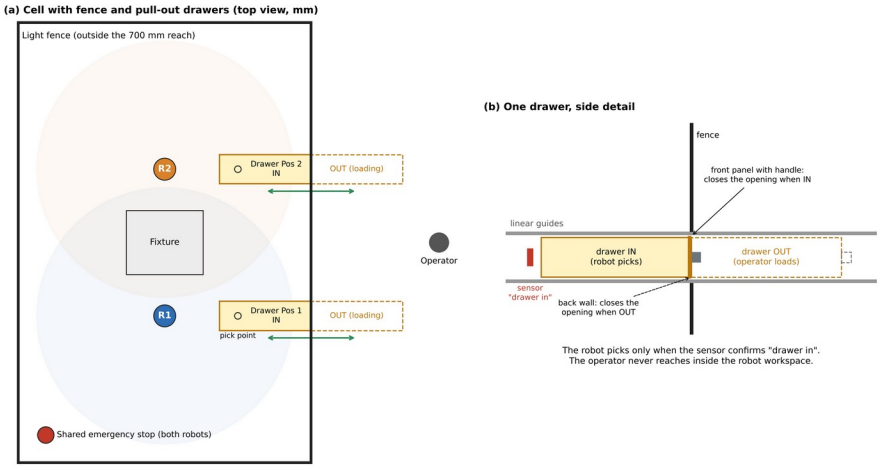


Figure 3.5 – Safety concept: light fence outside the reach and one pull-out drawer per robot.

4 Meeting the tolerances: the reality check

The xArm6 repeatability is ±0.1 mm (Hardware Manual V2.6.0), twice the ±0.05 mm required by the drawing. So the robot cannot set the final position of the parts. My answer has three parts: the robots only transport, the fixture locates, and the parts are measured and paired before loading.

4.1 Principle: decoupling accuracy from the robot

I do not try to make the robot more accurate. I remove it from the tolerance chain.

1. The robot places the part into the lead-ins of the fixture.
2. It opens the gripper and moves up.
3. Only then does the fixture centre, push and clamp the part.

After the release, the robot error no longer affects the result. The robot provides transport; the fixture provides accuracy.

This works under one condition: the arrival error must stay inside the capture range of the lead-ins. The arrival error is the robot repeatability (±0.1 mm) plus the tray error along the profile (about ±1 mm, assumed). I assume lead-ins with a capture range of ±2 mm; the exact value is a design parameter of the fixture.

Releasing before centring also avoids an over-constrained system, where the robot and the fixture push the part to two different positions. There is only one "owner" of the position: the fixture.

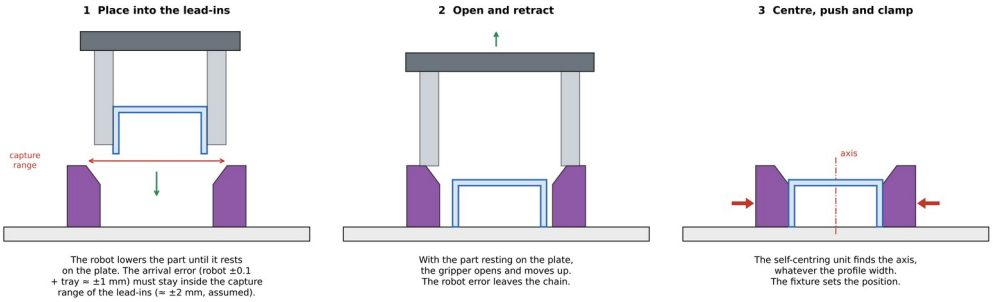


Figure 4.1 – Decoupling the accuracy from the robot: place, release, then centre. The fingers grip the profile at its centre, the centring units act near its ends.

4.2 Fixture design

Each dimension of the drawing is set by one fixture element. The rule for every part is: position first, then clamp.

Requirement (drawing)	Fixture element	Why this solution
120 ±0.05 axis to axis, Pos 1	Two self-centring pneumatic units per Pos 1, near the ends	The dimension is between axes. With a side stop, the width tolerance (40 ±0.3) would move the axes; the centring unit finds the axis whatever the width.
2 × 100 ±0.05, Pos 1 end to Pos 2 axis	Common end stop with lead-in for both Pos 1, pneumatic pusher from the opposite end	One reference for both parts. The pusher removes the tray error along the profile.
2 × 60 ±0.05, Pos 1 outer edge to Pos 2 end	Spring-loaded 60 mm gauge block pushed against the real Pos 1 edge, pusher on Pos 2	The centring moves the Pos 1 edge by up to ±0.15 mm (half the width tolerance). A fixed stop would not follow it.
120 ±0.05 axis to axis, Pos 2, and ⊥ 0.05	Two centring units per Pos 2, in the overhangs, about 220 mm apart	Wide spacing reduces the angular error: angle error ≈ point error / spacing.
// 0.05 between the two Pos 1	Centring units of each Pos 1 placed far apart	Same reason as above.
Gap ≤ 0.05 at the 4 contacts	Pneumatic swing clamps above the 4 crossings, with open/closed sensors	Pos 2 always rests on two supports, and the clamps press it down. Moderate force, because the sheet is only 2 mm thick.

The fixture holds the parts until joining, which is outside the robot task, and leaves access to the four crossings.

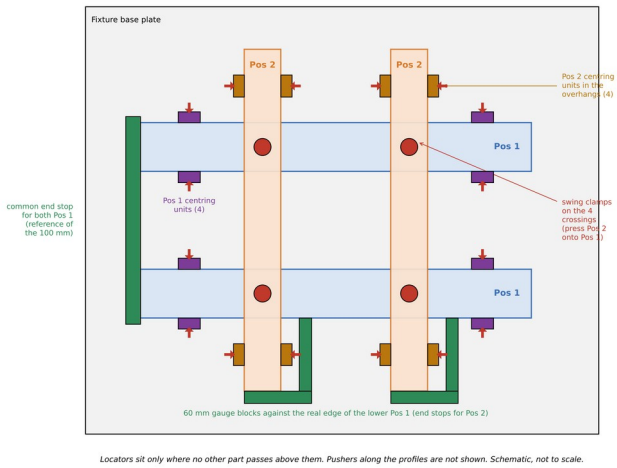


Figure 4.2 – Fixture concept (top view, schematic).

4.3 Error budget

I combine the fixture errors with a mixed method. Systematic errors, such as machining errors of the fixture, are measured once and corrected with adjustable locators. Random errors, such as the repeatability of the centring units and temperature, are combined by root sum of squares (RSS).

Example for the dimension 120 ±0.05 between the Pos 1 axes. The component values are assumptions, to be replaced with catalogue data:

Contributor	Value (mm, assumed)
Position of centring unit A (fixture machining)	±0.010
Position of centring unit B (fixture machining)	±0.010
Repeatability of centring unit A	±0.010
Repeatability of centring unit B	±0.010
Thermal expansion (aluminium fixture, steel part, ΔT = 5 K)	±0.007
Worst case (sum)	**±0.047**
RSS	**±0.021**

I give both values. The worst case assumes that all errors reach their maximum in the same direction at the same time. It is an upper limit: it is still inside ±0.05 mm, but with almost no margin. The RSS assumes independent random errors and is the realistic estimate (±0.021 mm). Both values still include the machining errors. Since I measure the fixture and correct these errors, the real result should be at or below the RSS value.

4.4 Part variation and open points

Parallelism of the Pos 2 top face. Pos 2 rests on both Pos 1. If the two Pos 1 have different heights (Δh), Pos 2 tilts. Over its length the tilt grows by 280 / 120 ≈ 2.3. To keep the parallelism within 0.05 mm, I need $\Delta h \lesssim 0.02$ mm (estimate for rigid, flat parts). The part height 20 has ±0.2 mm (ISO 2768-m), ten times more. The fixture cannot correct this.

My solution: before loading, the operator measures the part heights with a dial indicator on a surface plate. Then the operator loads a matched kit into the trays, with the two Pos 1 within 0.02 mm. I also measure the Pos 2 heights. The gap ≤ 0.05 is not a problem: the clamps press Pos 2 onto both supports.

Other options, in case kit matching is not enough: machining the top faces after joining (common in production), or changing the drawing.

Reading of "2 ×". I assume that 2 × 100 means 100 mm on each Pos 1, measured from the common left end. If it means 100 mm from both ends of the same part, the chain 100 + 120 + 100 = 320 needs the part length within ±0.15 mm, but ISO 2768-m gives ±0.5 mm. Then the parts must be selected by length or their ends machined. I sent this question to ETH.

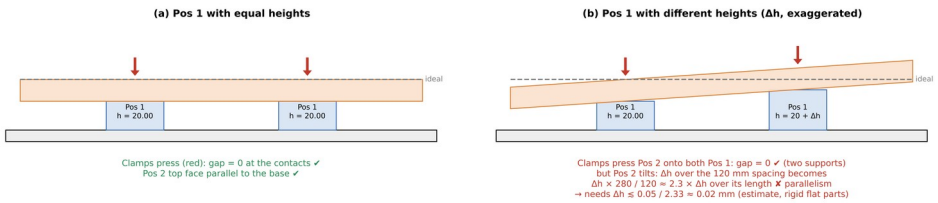


Figure 4.3 – Side view: a height difference between the two Pos 1 tilts Pos 2.

4.5 Validation

The matrix links each requirement to how the cell achieves it and how I check it.

Requirement	Achieved by	Verified by
120 axis to axis, Pos 1	Self-centring units	Fixture measured once; first assemblies fully measured
120 axis to axis, Pos 2	Centring units in the overhangs	Fixture measured once; first assemblies fully measured
2 × 100	Common end stop, pusher, Pos 2 centring units at 100 and 220 mm	Fixture measured once; first assemblies fully measured
2 × 60	Spring-loaded gauge block on the real Pos 1 edge	First assemblies fully measured
// and ⊥ 0.05	Wide spacing of the centring units	Fixture measured once; first assemblies fully measured
// 0.05, Pos 2 top face	Height-matched kits (Δh ≤ 0.02 mm)	Kit measured before loading; first assemblies fully measured
Gap ≤ 0.05	Swing clamps at the 4 crossings	0.05 mm feeler gauge on every assembly

I plan three checks:

1. **Fixture acceptance, once before use.** I measure the position of every locator (centring units, end stops, gauge block) and adjust it until its error is below the value assumed in the error budget. This removes the systematic errors.
2. **First assemblies, fully measured.** I measure all drawing dimensions on the first assemblies. If they are inside the tolerances, the cell and the error budget are confirmed. If not, I correct the fixture and repeat.
3. **Every assembly, quick check.** Before joining, the operator checks the four crossings with a 0.05 mm feeler gauge. If the gauge enters, the assembly is not joined and is checked.

Measuring ±0.05 mm needs instruments much more precise than the tolerance. A standard caliper is not enough; I use a dial indicator or the measuring machine of the lab.

5 Limitations and next steps

This is a concept: nothing has been built or tested yet.

- The values of the fixture components (centring units, lead-ins), the tool length and the tray accuracy are assumed.
- The reachability check covers geometry only, without collisions or dynamics.
- The cycle time is an estimate.

The next steps are:

1. Detailed fixture design in CAD, with catalogue components and a new error budget based on their data.
2. 3D simulation of the cell with collision checks.
3. Building the fixture, measuring it and running the first assemblies, as planned in Section 4.5.
4. For series production: splitting the work by side, higher speeds and a rotary table for loading.

6 Conclusion

Two xArm6 robots can build this frame if the accuracy is moved from the robots to the fixture. The robots place the parts into lead-ins and release them; the fixture centres, pushes and clamps them. The fixture cannot correct the height differences between the parts, which would tilt Pos 2. So the parts are measured and paired into kits before loading.

My error budget gives ±0.021 mm (RSS) and ±0.047 mm (worst case) against the required ±0.05 mm. The concept is simple enough for a lab batch and can be validated with the three checks of Section 4.5.

Appendix

A Reachability check: radial scan

Output of reach_check.py: gripper pointing down, TCP 60 mm above the base plane, tool length 200 mm (assumed). "ok" = inside my design band of 30–75% arm extension.

Distance from base (mm)	150	200	250	300	350	400	450	500	550	600	650	700
Arm extension (%)	18	24	31	38	45	52	60	68	75	83	91	98
Result	check	check	ok	ok	ok	ok	ok	ok	ok	check	check	check

B Cycle time estimate

Assumptions: transfer at 250 mm/s on straight lines; pick (approach, close, retreat) 3 s; place (approach, open, retreat) 3 s; locating (centring and pushing) 1 s; clamping 1 s. Tray-to-fixture distances from the layout: 525 mm (near point) and 610 mm (far point). R2 picks its first part during the R1 phase. The fixture locates the first part of each robot while the robot returns to the tray, so only the last locating step adds time.

Step	R1 phase (s)	R2 phase (s)
Pick first part	3.0	– (done in parallel)
Transfer to near point (525 mm)	2.1	2.1
Place	3.0	3.0
Return to tray	2.1	2.1
Pick second part	3.0	3.0
Transfer to far point (610 mm)	2.4	2.4
Place	3.0	3.0
Locate (and clamp, Pos 2) last part	1.0	2.0
Total	**19.6**	**17.6**

Total per assembly: about 37 s, rounded up to 40 s.

C References

- ETH Zurich, drawing AMZ-SA-26-001 (assignment material).
- UFACTORY, xArm Hardware Manual, model 1305, V2.6.0.
- UFACTORY, Kinematic and Dynamic Parameters – xArm 6.
- UFACTORY, xarm_ros2 repository, file xarm6_default_kinematics.yaml.
- ISO 2768-1, General tolerances.
- ISO 10218-1/-2, Robotics – Safety requirements for industrial robots; ISO/TS 15066, Collaborative robots.
- IEC 61131-3, Programmable controllers – Programming languages.
- EN 10162, Cold rolled steel sections.

D Supporting material and use of AI tools

GitHub repository: github.com/davide-91/eth-xarm6-assignment. It contains reach_check.py, the SoMachine project of the SFC, the PIPE model of the Petri net with its invariant analysis, and the figures.

AI tools were used for drafting support and code assistance; the concept, design decisions and verification are my own.